

DevOps Lab Program- 3

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Project Structure:

program-3/

→ Dockerfile

→ requirements.txt

→ app.py

Steps for execution:

Step 1: Create a **Dockerfile**

```
GNU nano 7.2 Dockerfile
FROM python:3.9-slim

LABEL version="1.0"

LABEL maintainer="aditin.mca24@rvce.edu.in"

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000
CMD ["flask", "run", "--port=5000", "--host=0.0.0.0"]
```

Step 2: Create a python file called **app.py**

```

GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")

def hello():
    return "Hi, docker!"

if __name__ == "__main__":
    app.run(port=5000,host="0.0.0.0")

```

Step 3: Create a file called **requirements.txt** with the following content

```

GNU nano 7.2 requirements.txt
flask

```

Step 4: Execute the docker build command on the terminal. Verify that the image is present once it has been built

```

lrv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-3$ docker build -t flask_app .
[+] Building 54.9s (11/11) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default
=> => transferring dockerfile: 367B                                              0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim                0.0s
=> [auth] library/python:pull token for registry-1.docker.io                    3.5s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 2B                                                  0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b 31.7s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b 0.1s
=> => sha256:38513bd7256313495cdd83b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB 25.4s
=> => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB 4.2s

```

```

o1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-3$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
flask_app latest 8be2d2760817 About a minute ago 133MB

```

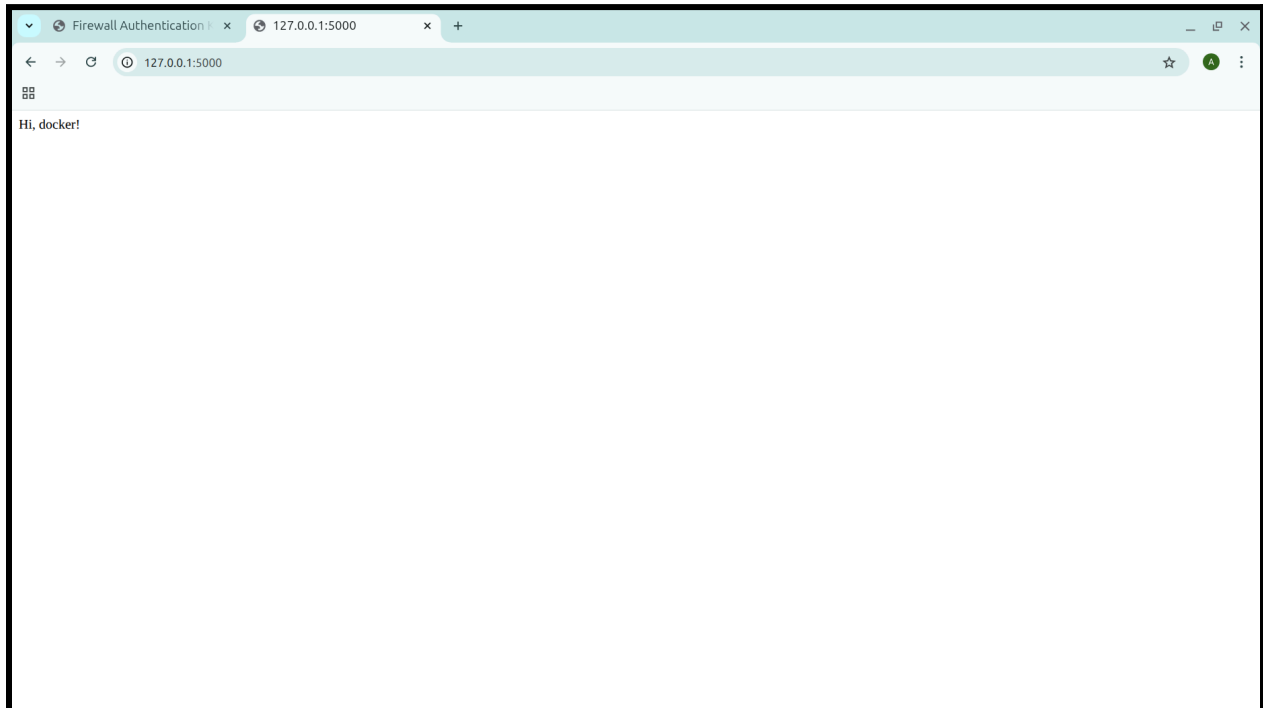
Step 5: Run the Docker Run command specifying the port number:

```

1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-3$ docker run -p 5000:5000 flask_app
* Serving Flask app 'app.py'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit

```

Step 6: Test the container by verifying the localhost details on the web browser.
The text “Hi, docker!” will be displayed.



Step 7: Stop and remove the docker container, then delete the docker image using its name

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-3$ docker container ls -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS          NAMES
a7fedac868ea   flask_app "flask run --port=50..." 3 minutes ago  Exited (0)    10 seconds ago          friendly_buck
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-3$ docker rm a7f
a7f
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-3$ docker rmi flask_app:latest
Untagged: flask_app:latest
Deleted: sha256:8be2d2760817095f4f588f7cb1a932909a6cb5d75869a6c9378b8546ca42d455
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-3$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
-
```